

Occam's Razor is Only as Sharp as Your ELBO

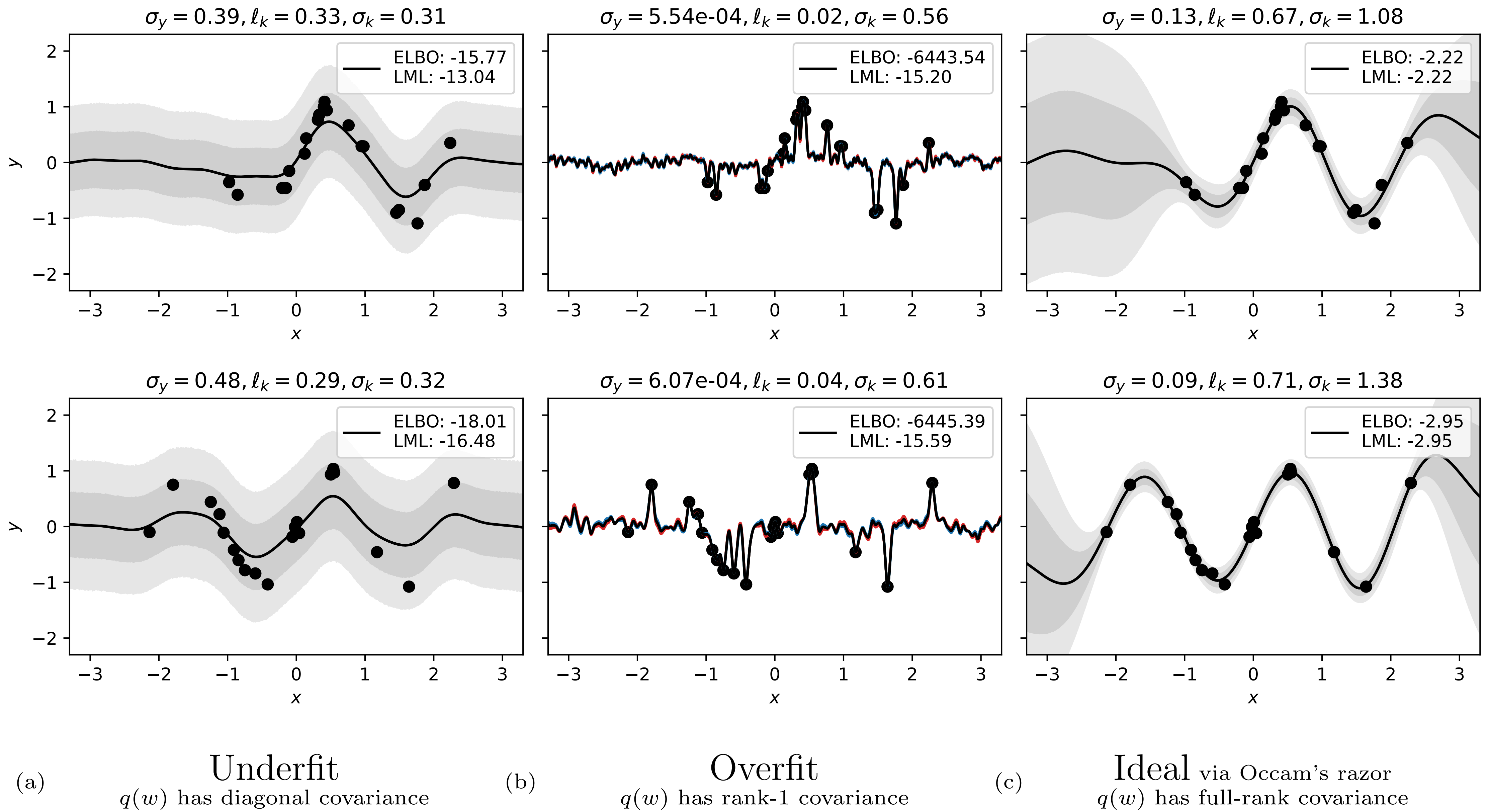


Figure 1: Predictive posterior for diagonal, rank-1, and full-rank covariance (columns) on different datasets of $N = 20$ inputs (rows). The ELBO can under or overfit based off your assumed approximate posterior.

BACKGROUND: REGRESSION

We have a training set $\{(x_i, y_i)\}_{i=1}^N$ where $x_i \in \mathbb{R}^D$ denotes an input vector and $y_i \in \mathbb{R}^D$ denotes a scalar output or target. Let $\phi(\cdot) : \mathbb{R}^D \rightarrow \mathbb{R}^R$ be a feature map that defines the design matrix $\Phi = [\phi(x_1)^\top; \dots; \phi(x_N)^\top] \in \mathbb{R}^{N \times R}$ and $y = [y_1; \dots; y_N] \in \mathbb{R}^N$ denote the target vector.

Prior and Likelihood.

$$p_\eta(w) = \mathcal{N}(w \mid 0_R, \Sigma_p) \quad (1)$$

$$p_\eta(y \mid w) = \mathcal{N}(y \mid \Phi w, \sigma_y^2 I_N) \quad (2)$$

Posterior. We have

$$p_\eta(w \mid y) = \mathcal{N}(w \mid \mu^*, \Sigma^*), \quad (3)$$

with $\mu^* = \frac{1}{\sigma_y^2} \Sigma^* \Phi^\top y$ and $\Sigma^* = \left(\Sigma_p^{-1} + \frac{1}{\sigma_y^2} \Phi^\top \Phi \right)^{-1}$.

Marginal Likelihood.

$$p_\eta(y) = \mathcal{N}(y \mid 0_N, \sigma_y^2 I_N + \Phi \Sigma_p \Phi^\top) \quad (4)$$

Approximate Posterior. Variational inference approximates the true (intractable) posterior $p(w \mid y)$ with a simpler distribution $q(w)$. We assume $q(w)$ is Gaussian, $q(w) = \mathcal{N}(w \mid \mu_q, \Sigma_q)$. We later explore different versions of the parametric form of the covariance matrix Σ_q , such as diagonal or rank-1.

“Note that we are at liberty to use as rich a family of approximating distributions as we please (there is no ‘overfitting’) and we are limited only by computational resources and the requirement of analytical tractability in the evaluation of the lower bound” (Barber and Bishop, 1998).

OVERFITTING EXAMPLE

Rank-1 Covariance. Consider restricting $q(w)$ to a rank-1 covariance matrix,

$$q(w) = \mathcal{N}(\mu_q, v_q v_q^\top + \varepsilon I_R), \quad (5)$$

where εI_R is a fixed small diagonal component added to keep the covariance positive definite.

Optimizing the ELBO leads to a null-space collapse.

The gradient here is

$$\frac{\partial J}{\partial v_q} = -\frac{1}{\sigma_y^2} \Phi^\top \Phi v_q - v_q + \frac{1}{\|v_q\|_2^2 + \varepsilon} v_q. \quad (6)$$

The objective for the mean parameters μ_q degenerates to a MAP-like objective, $J(\mu_q, \dots, \eta) := \text{const} +$

$$-\frac{1}{2} \left[N \log(2\pi\sigma_y^2) + \frac{1}{\sigma_y^2} \|y - \Phi \mu_q\|_2^2 + \|\mu_q\|_2^2 \right]. \quad (7)$$

As a result, the objective function with respect to the hyperparameters η becomes dominated by the log-determinant (first term above), which drives $\sigma_y \rightarrow 0$, as discussed in Bishop and Qazaz (1996).